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RESEARCH ARTICLE

Cannabis containing THC impairs 20-min cycling time trial performance irrespective of the method of inhalation

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Abstract

Herein, we examine the human exercise response following cannabis inhalation, taking into consideration varied cannabinoid concentrations and different inhalation methods. A semirandomized crossover study design was used, with measures of perceived exertion and physiological responses to submaximal and maximal exercise. Participants ($n = 14$, 9 males 5 females) completed exercise after *1*) smoking Δ-9-tetrahydrocannabinol (THC)-predominant cannabis (S-THC), *2*) inhaling aerosol (vaporizing) from THC-predominant cannabis (V-THC), *3*) inhaling aerosol from cannabidiol (CBD)-predominant cannabis (V-CBD), or *4*) under control conditions. All exercise was performed on a cycle ergometer, with submaximal testing performed at 100 W followed by an evaluation of maximal exercise performance using an all-out 20-min time trial. Metabolism was characterized via the analysis of expired gases while subjective ratings of perceived exertion (RPE) were reported. During submaximal cycling, heart rate was higher during S-THC and V-THC compared with both control and V-CBD (all $P < 0.02$). During maximal exercise, $\dot{V}_E$ was lower in V-THC compared with control, S-THC, and V-CBD (all $P < 0.03$), as was S-THC compared with control ($P < 0.05$). Both $\dot{V}O_2$ and RPE were similar between conditions during maximal exercise (both $P > 0.1$). Mean power output during the 20-min time trial was significantly lower in the S-THC and V-THC conditions compared with both control and V-CBD (all $P < 0.04$). Cannabis containing THC alters the physiological response to maximal and submaximal exercise, largely independent of the inhalation method. THC-containing cannabis negatively impacts vigorous exercise performance during a sustained 20-min effort, likely due to physiological and psychotropic effects. Inhalation of cannabis devoid of THC and primarily containing CBD has little physiological effect on the exercise response or performance.

NEW & NOTEWORTHY Inhalation of cannabis containing THC alters physiological responses to both submaximal and maximal exercise and reduces mean power output during a 20-min time trial, regardless of whether it is inhaled as smoke or aerosol. In contrast, cannabis devoid of THC and predominantly containing CBD has no effect on physiological responses to exercise or performance.

cannabinoids; cycling performance; marijuana; smoking; vaporizing

INTRODUCTION

Cannabis is the second most commonly used recreational substance in the world, with an estimated 200 million users (1). In many regions, there is an increasing acceptance of cannabis use—both socially and legally (2–4), suggesting that recreational and medicinal use will only continue to become more commonplace. This includes the consumption of both whole cannabis and its unique constituent molecules, referred to as cannabinoids. Evidence suggests that within the larger population, athletes across the spectrum of sports and competition levels (5–7) are turning to cannabis for a variety of purposes, including purported performance enhancement (6, 8, 9), aid in recovery (10, 11), and recreation (6, 8). The use of cannabis and cannabinoids by athletes is highly controversial, given longstanding concerns regarding athlete health and/or fair play.

The scientific literature focusing on cannabis and human exercise performance is presently incomplete, owing both to a dearth of studies and documented limitations (5, 12–15). All the same, existing works still offer select insight into the effects of cannabis use on human physiology and possibly performance. Seminal investigations examining cannabis use in clinical populations who performed exercise report an acceleration of the time to failure in a graded exercise test, which coincided with the onset of angina in patients with coronary heart disease (16, 17). Subsequent studies in healthy volunteers consistently demonstrated cannabis use to result in an elevated heart rate for a given submaximal workload (18, 19), preceding the assumption that the maximal work rate would also be reduced (19). Follow-up work highlighted that although cannabis elevated heart rate during submaximal exercise, this effect disappeared as work rate progressed toward maximal intensity (20). Even so, time to exhaustion on an incremental cycling test was still reduced (20).

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Collectively, these findings support the general belief that cannabis inhalation is ergolytic (5, 14, 21, 22); however, this conclusion may be erroneous for a number of reasons. First, much of the existing evidence is drawn from studies in clinical populations (16, 17, 23), whose ability to sustain high work rates differs substantially from that of athletes (24), or even recreationally active populations. Given that highly trained and lesser trained individuals are physiologically limited by different factors (25), the generalizability of these findings across groups may be limited. In addition, although elevated heart rate has repeatedly been demonstrated during submaximal exercise following cannabis use, there are numerous examples of pharmacological/nutritional supplements that similarly increase heart rate and yet are clearly ergogenic, such as caffeine and amphetamines (26, 27). This brings into question the supposition that cannabis reduces maximal performance based on its effects on submaximal heart rate per se. Perhaps most importantly, data during maximal exercise following cannabis inhalation are remarkably scarce, although limited evidence suggests that time to exhaustion during graded cycling exercise may be reduced (20). Although an ability to endure increasing exercise intensities could be important for performance in select sports, the ecological validity of this test is questionable in other typical scenarios, such as a race wherein the objective is to complete a set distance in the shortest possible time. As such, time to exhaustion provides limited information on an athlete's real-world performance capabilities (28). Although more ecologically valid performance tests have recently been trialed following the consumption of edible cannabis products (29, 30), no investigation has sought to examine inhalation.

Commercially available cannabis products can contain high concentrations of the sought-after psychoactive and tachycardia-inducing cannabinoid Δ-9-tetrahydrocannabinol (THC) (31). There is also growing interest in the use of the nonintoxicating cannabinoid cannabidiol (CBD). Alongside changing cannabinoid concentrations, the methods of cannabis inhalation have also evolved, such that traditional "smoking" is not the only option. Cannabis is now commonly inhaled by aerosol, using a family of devices known as "dry flower vaporizers," which circumvent the necessity to inhale smoke and the by-products that come with the combustion of dried plant material, yet accelerate the pharmacodynamics of cannabinoids (32, 33). Given the prevalence of inhalation as a means to use cannabis, a study of both different forms of inhalation and the different types of cannabis products available to contemporary athletes is warranted. More broadly, to improve the scientific knowledge surrounding the many controversies of cannabis use in sports, targeted investigations that specifically seek to determine the effects of cannabinoids on the physiological response during exercise and outright performance are needed.

The purpose of this study was to examine the effects of both varied cannabinoid concentrations and cannabis inhalation methods on the physiological response and exercise performance of healthy athletic adults. We hypothesized that THC-predominant cannabis would have detrimental impacts on performance, possibly due to both its psychotropic effects on pacing and an alteration of cardiorespiratory function. On the contrary, we hypothesized that cannabis devoid of THC and predominantly containing CBD would have no effect on perceived exertion or physiological responses to exercise and, therefore, have no effect on performance. Finally, it was hypothesized that although both smoked and vaporized THC-predominant cannabis would impair the performance, the use of smoking as an inhalation method would be associated with greater decrements due to the rise in carboxyhemoglobin levels associated with smoke inhalation, despite the faster pharmacodynamics observed with vaporization.

MATERIALS AND METHODS

Participants

Participants who were habitual cannabis users (minimum use 1 time/wk for 1 mo; confirmed by a urine drug test) were recruited from the community and local club-level cycling and rowing teams. The only other inclusion criterion was that participants were 19 yr of age (the legal age for cannabis use in Canada). Exclusion criteria included current or past substance use disorder or mood/anxiety disorder, failure of urine screening test for use of other recreational drugs, a family history of psychosis or a previous cannabis-related psychotic episode, diagnosis of chronic disease, hypertension, pregnancy, and current use of prescription medication other than oral contraceptives. Each participant provided written informed consent before completing any study procedures. This study was completed in accordance with the Declaration of Helsinki and was part of a larger registered clinical trial (NCT04693884), which was approved by both Health Canada and the institutional research ethics board.

Procedure

Participants visited the laboratory for one introductory visit, followed by four experimental visits, which occurred in a semirandomized order determined by a randomization software (random.org). During the introductory visit, participants' height and weight were measured using a standard stadiometer (SECA, Hamburg, Germany) and a digital scale (Tanita, Tokyo, Japan), which was calibrated before each use. Aerobic capacity was measured during an incremental ramp test to exhaustion (20–30 W/min increase) on an electronically braked cycle ergometer (Velotron Quarq, Spearfish, SD) with breath-by-breath indirect calorimetry (COSMED, Rome, Italy) to quantify maximal oxygen uptake ($\dot{V}O_{2max}$). Maximal aerobic power was defined as the maximum power output achieved during the test.

For each experimental visit, participants arrived at the laboratory having abstained from cannabis use for ≥48 h, exercise, alcohol, and nutritional supplements for ≥24 h, caffeine for ≥8 h, and food for ≥6 h. All visits were conducted a minimum of 3 days apart. Upon arrival, participants provided a urine sample, which was analyzed for a THC metabolite (11-COOH-THC) using a commercial urine test strip with three distinct detection levels of 18, 50, and 150 ng/mL (NarcoCheck, Montluçon, France) to confirm ongoing (detection at 18 or 50 ng/mL) but not recent (detection at 150 ng/mL) cannabis use. The diet was not controlled before each visit; however, we requested that participants recreate similar meals on the days before each

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visit. The time of day at which visits occurred was not standardized between participants, nor were all visits conducted at identical times of day within participants. However, efforts were made to conduct visits either consistently in the morning or afternoon for a given participant. Each condition required a 30-min exercise bout on the cycle ergometer with 10 min of submaximal exercise at a power output of 100 W, followed by a 20-min time trial, with the explicit goal of achieving the highest attainable mean work rate through maximal effort. A 20-min time trial was selected based on its established relationship with functional threshold power (34). The test was performed under control conditions on the first visit and following one of three randomized cannabis inhalation protocols on subsequent visits. In each condition, the exercise was initiated <20 min from the start of cannabis inhalation to account for the rapid arrival of cannabinoids in the blood (35), during which time participants were also instrumented for heart rate and metabolic measures. Briefly, the three protocols involved smoking THC-predominant cannabis (S-THC), vaporizing THC-predominant cannabis (V-THC), or vaporizing CBD-predominant cannabis (V-CBD). Inhalation protocols are described in greater detail in the section entitled *Cannabis*. Control procedures were performed identically to those performed on visits wherein cannabis was inhaled, with the exception of the cannabis inhalation before exercise. Instead, participants were directed to the bike immediately after the initial procedures.

Indirect calorimetry was used throughout submaximal and maximal exercise. Ratings of perceived exertion (RPE) were recorded every 2 min during exercise using a 6–20 Borg scale (36). During the 20-min time trial, participants could control and see their selected gearing ratio, cadence, and elapsed time, but were blinded to distance completed, power output, and heart rate. No encouragement was provided during the 20-min time trial, and the only interaction between researchers and participants was the evaluation of RPE.

Cannabis

Cannabis inhalation was performed in a custom-built inhalation chamber (Supplemental Figure), which ensured adequate airflow and filtration for indoor smoke inhalation with a minimum of recirculating smoke. All cannabis was sourced from an accredited manufacturer (Aurora Cannabis Inc., Edmonton, AB, Canada) and was analytically tested to confirm exact cannabinoid concentrations. Three separate administration procedures were used, with each involving a unique combination of inhalation method and cannabis variant. Participants were instructed to follow an inhalation protocol involving a 5-s inhalation, followed by a 10-s hold, with 45 s in between inhalations. However, for safety and practicality, deviations from the protocol were permitted as necessary (e.g., coughing or laughing), and no set number of inhalations were required. The protocols were completed within 10 min. The inhalation methods were traditional smoking using a metal pipe and ignition source, and inhalation using a dry-cannabis flower vaporizer (Storz and Bickel, Tuttlingen, Germany) set to 190°C for aerosolization and activation of cannabinoids without combustion and smoke generation. Two forms of cannabis were used. The first contained higher levels of THC and had negligible CBD content (100 mg, 13.6% THC, <1% CBD); the second contained higher CBD and had negligible THC content (100 mg, 14.5% CBD, <1% THC). No protocol involving smoking of high-CBD cannabis was used in the present study, as the a priori research questions were to assess the independent effects of *1*) the inhalation method and *2*) the cannabinoid composition within inhaled cannabis and not the interactive effects of these two factors. Neither researchers nor participants were blinded to the administered treatments.

Data and Statistical Analysis

Means for experimental conditions were compared across groups and time using a linear mixed model, with condition and time as a fixed effect and individuals as a random effect. Post hoc comparisons were performed in the presence of a significant main effect or interaction when appropriate and were completed using Fisher's least-significant difference (LSD) tests. Submaximal and maximal exercise data were averaged throughout the entire test. Change scores for each condition are reported relative to the control condition. Data reported at discrete 2-min time points were determined by averaging all data points occurring within 10 s of the stated time. Ventilatory equivalents were calculated by dividing minute ventilation ($\dot{V}_E$) by oxygen consumption ($\dot{V}O_2$) and carbon dioxide production ($\dot{V}CO_2$). The oxygen pulse was calculated by dividing relative $\dot{V}O_2$ by heart rate. Gross efficiency (GE) was calculated during the submaximal exercise bout by dividing the total work completed during this period by the total energy cost, which was determined using the average $\dot{V}O_2$, the respiratory exchange ratio (RER), and standard values. Statistical significance was defined a priori as $P \leq 0.05$. All statistical analysis was completed using Prism 9 (San Diego, CA). Researchers were not blinded to the various conditions during data analysis.

RESULTS

Fifteen participants enrolled in the present study; however, one male participant contracted COVID-19 before completing all study visits and was subsequently excluded; thus, the final study sample included 14 participants (9 males, 5 females). All qualified as *tier 1* or *tier 2* athletes (37). Descriptive participant data are presented in Table 1. During submaximal cycling at 100 W, mean heart rate was higher during S-THC and V-THC compared with both control and V-CBD (ΔS-THC: 14 ± 19 beats/min, ΔV-THC: 18 ± 19 beats/min, ΔV-CBD: 1 ± 10 beats/min; all $P < 0.02$; Fig. 1*A*). During maximal exercise, no differences were apparent (ΔS-THC: 2 ± 13 beats/min, ΔV-THC: 0 ± 15 beats/min, V-CBD: −3 ± 9 beats/min; $P = 0.4$; Fig. 1*B*). When heart rate was examined at discrete 2-min time points, the difference in submaximal heart rate was also reflected, such that heart rate was greater at all submaximal exercise time points in S-THC and V-THC compared with control and V-CBD (all $P < 0.001$; Fig. 1*C*).

The cardiorespiratory outcomes measured throughout exercise are displayed in Table 2. $\dot{V}_E$ was lower in V-THC compared with S-THC and V-CBD during submaximal exercise (both $P < 0.03$). Similarly, during maximal exercise, $\dot{V}_E$ was lower in V-THC compared with control, S-THC, and

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Table 1. *Characteristics of participants who completed all study procedures, presented as a single group, and divided by sex*

	All	Male	Female
	n = 14	*n* = 9	*n* = 5
Age, yr	23 ± 4	25 ± 4	21 ± 2
Height, m	1.72 ± 0.05	1.74 ± 0.04	1.67 ± 0.04
Weight, kg	73.0 ± 8.2	76.7 ± 7.8	66.3 ± 2.5
BMI, kg·m^{-2}	24.7 ± 2.1	25.2 ± 2.2	23.7 ± 1.2
$\dot{V}O_{2max}$, L·min^{-1}	3.73 ± 0.82	4.14 ± 0.74	3.00 ± 0.19
$\dot{V}O_{2max}$, mL·kg^{-1}·min^{-1}	50.6 ± 7.3	53.6 ± 7.5	45.3 ± 2.3
Max aerobic power, W	319 ± 81	355 ± 79	254 ± 25
Weekly cannabis use (sessions/wk)			
1–2	12	7	5
3–4	2	2	0

Data are presented as means ± SD; *n* denotes number of participants. Maximal aerobic power represents the greatest power output achieved on the incremental $\dot{V}O_{2max}$ test. BMI, body mass index.

V-CBD (all $P < 0.03$), as was S-THC compared with control ($P < 0.05$). Respiratory frequency (f) was similar between conditions during submaximal ($P = 0.5$) and maximal exercise ($P = 0.3$). Tidal volume (V_t) differed between conditions, both during submaximal and maximal exercise. During submaximal exercise, V_t was lower in V-THC compared with control and V-CBD (both $P < 0.04$). During maximal exercise, V_t was lower in both S-THC and V-THC compared with both control and V-CBD (all $P < 0.03$). $\dot{V}CO_2$ was elevated during submaximal exercise in S-THC compared with V-THC and V-CBD (both, $P < 0.02$) but not control ($P = 0.1$). There was no difference between conditions in $\dot{V}CO_2$ during maximal exercise ($P = 0.4$). $\dot{V}O_2$ was higher during submaximal exercise in S-THC compared with both V-THC and V-CBD (both $P < 0.04$), but not control ($P > 0.05$; Table 2, Fig. 2*A*). Similar to $\dot{V}CO_2$, $\dot{V}O_2$ did not differ across conditions during maximal exercise, including fractional use of $\dot{V}O_{2max}$ (ΔS-THC: −0.5% ± 9.6%, V-THC: −3.2% ± 9.3%, V-CBD: −0.3% ± 5.0%; $P = 0.4$; Fig. 2*B*), or $\dot{V}O_2$ at any time point (all $P > 0.05$, Fig. 2*C*). There were no differences in ventilatory equivalents during submaximal exercise ($\dot{V}E/\dot{V}O_2$ or $\dot{V}E/\dot{V}CO_2$); however, during maximal exercise, both equivalents were reduced in S-THC and V-THC compared with both control and V-CBD (all $P < 0.01$). Despite submaximal differences in $\dot{V}CO_2$ and $\dot{V}O_2$, the respiratory exchange ratio (RER) was similar between conditions for both submaximal ($P = 0.7$) and maximal exercise ($P = 0.6$). O_2 pulse during submaximal exercise was reduced in S-THC compared with control ($P = 0.04$) and reduced in V-THC compared with all conditions (all $P < 0.0001$), but similar across conditions during maximal exercise. As a consequence of greater $\dot{V}O_2$ at a fixed submaximal workload, GE was reduced in S-THC compared with both V-THC and V-CBD (both $P < 0.03$). GE was also greater in V-CBD than in control ($P < 0.02$).

RPE was similar between conditions both during submaximal (ΔS-THC: 0.3 ± 1.6, ΔV-THC: 0.1 ± 1.4, V-CBD: −0.3 ± 1.0; $P = 0.35$; Fig. 3*A*) and maximal exercise (ΔS-THC: −0.2 ± 1.2, ΔV-THC: −0.4 ± 1.2, V-CBD: −0.9 ± 1.4; $P = 0.1$; Fig. 3*B*) and was similar at all individual time points throughout the exercise bout ($P = 0.9$; Fig. 3*C*) under all conditions.

Despite no differences in RPE throughout the 20-min time trial, mean power output was significantly lower in the S-THC and V-THC conditions compared with both control and V-CBD (all $P < 0.04$; Fig. 4, *A* and *B*). Power output expressed as a percentage of max aerobic power did not differ throughout the exercise, suggesting that the pacing strategy remained similar in each condition ($P = 0.2$; Fig. 4*C*).

DISCUSSION

This study demonstrates that the inhalation of cannabis containing THC alters physiological responses to both submaximal and maximal exercise and is associated with decreases in exercise performance. The effect on performance appears to occur irrespective of the method of inhalation. Unlike THC-predominant cannabis, inhaling CBD-predominant cannabis devoid of THC does not appear to alter physiological responses to exercise and consequently has no effect on exercise performance.

Herein, we report a substantial elevation in heart rate during submaximal exercise following the inhalation of cannabis containing THC, in agreement with earlier works (18–20).

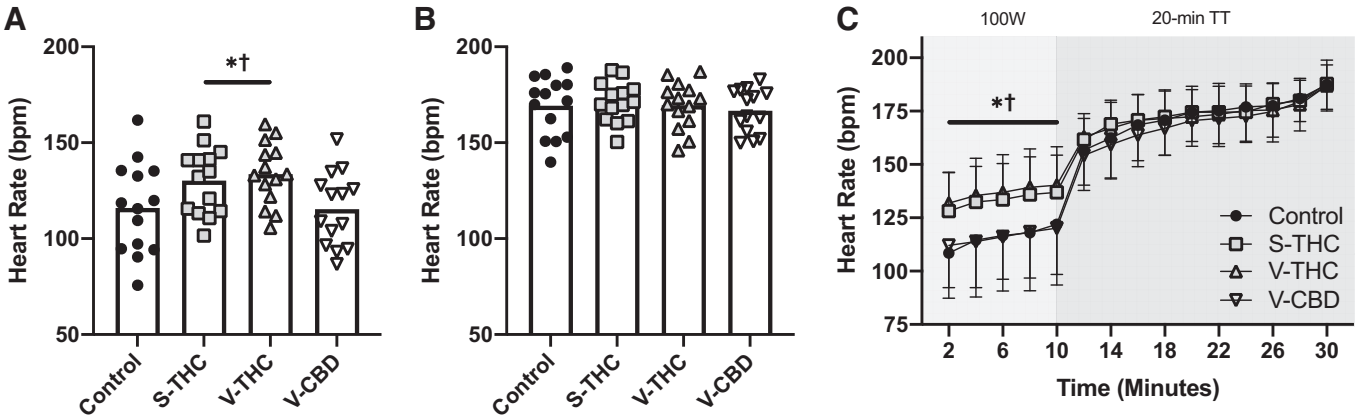

Figure 1. Mean heart rate of participants who completed a control condition and three varied conditions of cannabis consumption during submaximal exercise (*A*) and a 20-min time trial (*B*). Bars represent mean and symbols represent individual values. Heart rate is also displayed at discrete 2-min time points throughout both exercise periods (*C*). Symbols represent mean, and error bars represent SD. *$P < 0.05$ vs. control, †$P < 0.05$ vs. V-CBD.

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Table 2. *Physiological outcomes determined by indirect calorimetry during submaximal exercise and a 20-min time trial following control and cannabis conditions*

	Submaximal				20-Min TT			
	Control	S-THC	V-THC	V-CBD	Control	S-THC	V-THC	V-CBD
$\dot{V}_E$, L·min^{-1}	40.2 ± 6.2	41.0 ± 5.9#	37.8 ± 5.0†‡	40.3 ± 5.8#	102.7 ± 32.0	94.0 ± 28.6*#	87.4 ± 28.3*†‡	99.3 ± 28.8#
V_t, L	1.81 ± 0.31	1.71 ± 0.32	1.69 ± 0.27*†	1.77 ± 0.28#	2.41 ± 0.46	2.23 ± 0.42*†	2.18 ± 0.45*†	2.38 ± 0.54#‡
f, s^{-1}	23 ± 5	25 ± 5	23 ± 5	23 ± 5	42 ± 9	42 ± 8	40 ± 8	41 ± 6
$\dot{V}_{CO_2}$, L·min^{-1}	1.44 ± 0.24	1.55 ± 0.22#†	1.39 ± 0.19‡	1.42 ± 0.21‡	2.96 ± 0.72	3.04 ± 0.78	2.83 ± 0.80	2.95 ± 0.77
$\dot{V}_{O_2}$, L·min^{-1}	1.72 ± 0.24	1.82 ± 0.26#†	1.65 ± 0.21‡	1.70 ± 0.21‡	3.07 ± 0.72	3.15 ± 0.82	2.98 ± 0.84	3.10 ± 0.81
$\dot{V}_E/\dot{V}_{O_2}$	23.3 ± 1.7	22.6 ± 2.8	22.9 ± 2.4	23.7 ± 2.2	33.1 ± 4.0	29.9 ± 4.4*†	29.3 ± 4.2*†	32.0 ± 3.6#‡
$\dot{V}_E/\dot{V}_{CO_2}$	28.0 ± 2.3	26.6 ± 3.0	27.3 ± 3.0	28.6 ± 2.7	34.4 ± 3.7	31.0 ± 4.7*†	30.9 ± 4.4*†	33.7 ± 3.6#‡
RER	0.83 ± 0.04	0.85 ± 0.07	0.84 ± 0.04	0.83 ± 0.06	0.96 ± 0.03	0.96 ± 0.08	0.95 ± 0.03	0.95 ± 0.04
$\dot{V}_{O_2}$, mL·kg^{-1}·min^{-1}	23.8 ± 4.1	24.8 ± 3.6#†	22.8 ± 3.3‡	23.4 ± 3.5‡	41.7 ± 6.0	42.3 ± 8.3	40.2 ± 7.4	41.8 ± 7.0
O_2 pulse, mL·kg^{-1}·beat^{-1}	0.21 ± 0.04	0.20 ± 0.03*#	0.17 ± 0.03*†‡	0.20 ± 0.04#	0.25 ± 0.04	0.25 ± 0.05	0.24 ± 0.04	0.25 ± 0.05
GE, %	15.9 ± 2.0	15.7 ± 2.2#†	16.5 ± 2.1‡	16.8 ± 1.7*‡				

Data are presented as means ± SD. Fisher's least significant difference post hoc comparisons were performed to identify between group differences only in the presence of a significant fixed effect of condition. f, respiratory frequency; GE, gross efficiency; RER, respiratory exchange ratio; TT, time trial; $\dot{V}_E$, ventilation, V_t, tidal volume; $\dot{V}_{CO_2}$, carbon dioxide production; $\dot{V}_{O_2}$: oxygen consumption. *$P < 0.05$ vs. control; ‡ vs. S-THC; # vs. V-THC, † vs. V-CBD.

The difference between the THC and control conditions in this study ranged from 14 to 18 beats/min, which is of a similar magnitude to that reported at rest (38) and during fixed workload cycling across populations (18, 20). The specific driver of this tachycardiac response remains to be firmly established, but it has generally been accepted to follow a global sympathetic nervous system activation (39), and tachycardia can be prevented with β-adrenoreceptor blockade (40). However, increased activity may be specific only to cardiac sympathetic nerves, as sympathetic activity targeting the vasculature is actually decreased following cannabis inhalation (41), which may have further implications for the regulation of blood flow during exercise (42).

Although stroke volume was not measured in the present study, this represents an important area for future investigation to disentangle the effects on cardiac output. It may be that the increased exercising heart rate resulted in an exaggerated cardiac output at a given submaximal workload, contributing to the higher $\dot{V}_{O_2}$ and reduced GE observed after smoking cannabis. Alternatively, the increased heart rate at a fixed workload could be interpreted to suggest a reduction in stroke volume during exercise if cardiac output remains matched to the demands of the exercise. Although the acute cardiac effects of cannabis inhalation are not well characterized, it has been reported that cannabis inhalation can reduce stroke index at rest (43). It is noted that in the present study, heart rate was increased to a similar magnitude in the V-THC condition, but $\dot{V}_{O_2}$ and GE remained unaffected. Thus, future work clarifying whether or not cardiac function is impaired during exercise by cannabis inhalation is warranted.

We observed a reduction in average power during 20 min of maximal exercise when cannabis contained THC, regardless of the method of inhalation. Despite reduced output, the physiological response during exercise remained largely similar across conditions, with the exception of a reduction in V_t and consequently $\dot{V}_E$ in all THC conditions. Although this reduction in $\dot{V}_E$ can be interpreted as obligatory to reduced work output in these conditions, it is notable that other parameters reflective of metabolic demand ($\dot{V}_{O_2}$, $\dot{V}_{CO_2}$, and RER) remained unchanged. This is pertinent given previous work identifying a decreased ventilatory sensitivity to CO_2

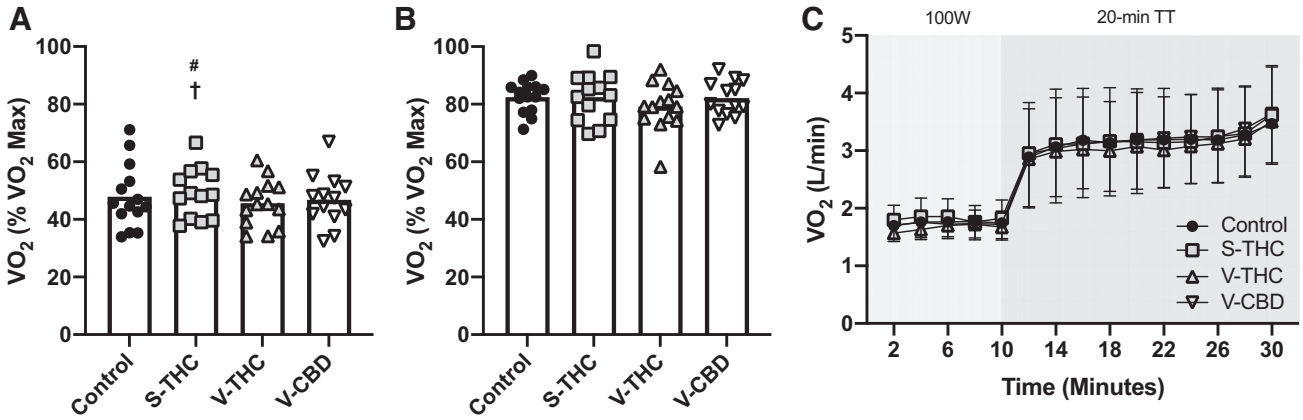

Figure 2. Mean $\dot{V}_{O_2}$ expressed as a percentage of $\dot{V}_{O_{2max}}$ during submaximal exercise (*A*) and a 20-min time trial (*B*) after control and three separate cannabis conditions. Bars represent mean and symbols represent individual values. $\dot{V}_{O_2}$ is also displayed at discrete 2-min time points throughout both exercise periods (*C*). Symbols represent mean, and error bars represent SD (*C*). Fisher's least significant difference post hoc comparisons were performed to identify between group differences only in the presence of a significant fixed effect of condition for *A* and *B*. For *C*, they were only performed in the presence of a significant interaction effect between time and condition. #$P < 0.05$ vs. V-THC, †$P < 0.05$ vs. V-CBD.

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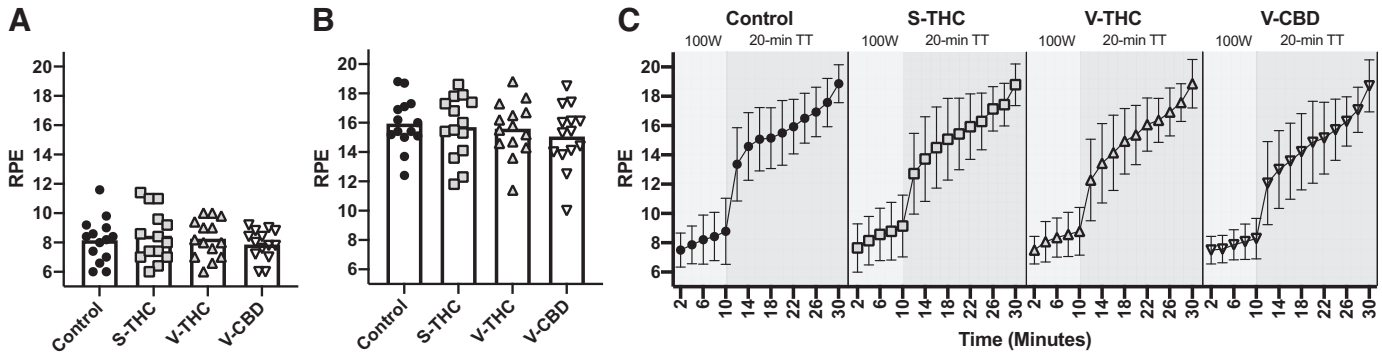

Figure 3. RPE reported on a 6-20 scale during submaximal exercise (*A*) and a 20-min time trial (*B*) following control, and three separate cannabis conditions. Bars represent mean and symbols represent individual values. RPE is also displayed at discrete 2-min time points throughout both exercise periods (*C*). Fisher's least significant difference post hoc comparisons were performed to identify between group differences only in the presence of a significant fixed effect of condition for *A* and *B*. For *C*, they were not performed due to the absence a significant interaction effect between time and condition. Symbols represent mean, and error bars represent SD. RPE, ratings of perceived exertion.

following THC administration at rest (44). These largely similar physiological responses to maximal exercise concomitant to reductions in work output suggest that inhalation of THC-containing cannabis introduces some degree of physiological inefficiency, which is collectively supported by an elevated heart rate and reduction in GE at a submaximal exercise intensity, and depression in V_t and $\dot{V}E$ during maximal exercise. The present methods cannot specifically pinpoint the step at which the translation of whole-body metabolic activity to external work output may meet interference.

Our data demonstrating reduced performance and altered physiological responses with maximal exercise contrast a recent examination of the effects of edible THC-containing cannabis products on cycling performance (29). Although notably different in the method of cannabis administration, Ewell et al. demonstrated that power output during both an all-out sprint test and a 20-min cycling time trial identical to that used in the present study was not impacted by THC ingestion. Although this study elegantly demonstrated increases in circulating THC metabolite concentrations concomitant with subjective indicators of drug effects at the time of exercise testing, heart rate was not elevated by THC at any time point during the 20-min time trial or during submaximal exercise. Thus, it may be that inhalation may be requisite for THC to interfere with cardiorespiratory responses to exercise and exercise capacity; both effects may then be circumvented by ingestion. Recent work has demonstrated that some of the cardiovascular effects of cigarette smoke are driven by vagal input from the airways (45), supporting the principle that inhalation itself may be required to facilitate these effects of THC. The effects of inhalation may be even more nuanced considering the differences we observed between smoked and vaporized cannabis in the present study. Although both S-THC and V-THC conditions demonstrated reduced performance and mostly similar physiological effects, it is interesting to note some of the latter were accentuated in V-THC, for example, the depression $\dot{V}E$ and O_2 pulse. These findings align with data suggesting that vaporizing THC leads to greater and faster pharmacodynamic effects than smoked THC (33, 35), though we cannot confirm this to be the case in the present study given that we did not measure blood cannabinoid levels.

Unlike THC-predominant cannabis, inhalation of CBD-predominant cannabis appeared to result in minimal differences in the physiological response to exercise, with the only exception being improved GE at a submaximal exercise

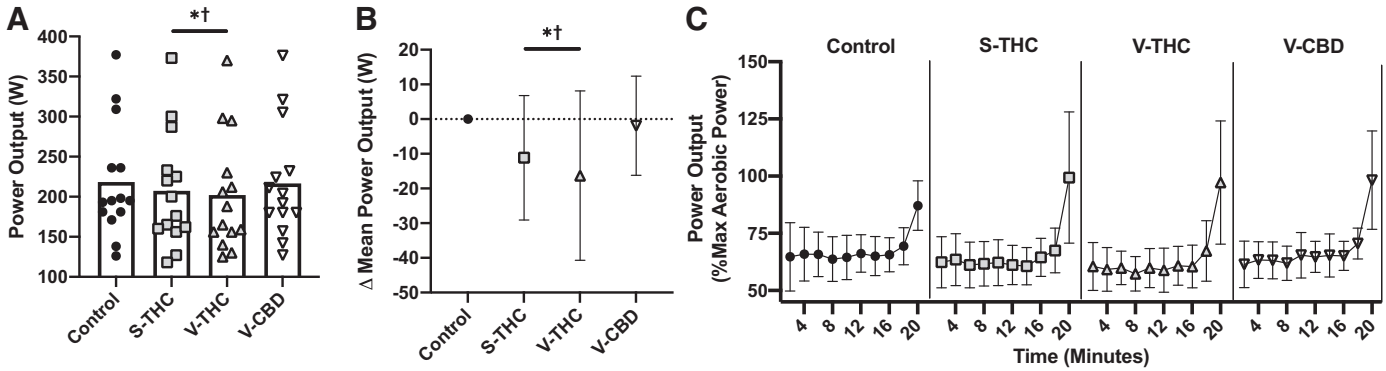

Figure 4. Power output during the 20-min time trial expressed in absolute terms (*A*) and as a change score from control (*B*), following three separate cannabis conditions. Bars represent mean and symbols represent individual values. Change score symbols represent mean and error bars represent SD. Power output is also expressed at 2-min time points throughout the 20-min time trial as a percentage of maximal aerobic power (*C*). Fisher's least significant difference post hoc comparisons were performed to identify between group differences only in the presence of a significant fixed effect of condition for *A* and *B*. For *C*, they were not performed due to the absence a significant interaction effect between time and condition. Symbols represent mean, and error bars represent SD. *$P < 0.05$ vs. control, †$P < 0.05$ vs. V-CBD.

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intensity compared with control conditions and when THC-predominant cannabis was smoked. This improved efficiency did not translate to any benefits during maximal exercise, as there was no difference in mean 20-min time trial power output in V-CBD compared with control. This finding contrasts previous work demonstrating that CBD increases $\dot{V}O_2$ at submaximal intensity and $\dot{V}O_{2max}$ (30). The reason for these discrepant findings is not immediately clear; however, it should be noted that the method of administration of CBD (oral vs. aerosolized whole cannabis), dose administered (300 vs. 14.6 mg), and exercise test (incremental run to exhaustion vs. 20-min cycling time trial) differed between studies. Ultimately, both investigations demonstrate no effect of CBD on specific exercise performance outcomes (30).

Alongside the physiological alterations that could contribute to reductions in power output, the psychoactive effects of THC and the potential effects on exercise performance must be considered. THC is a cannabinoid with well-established psychotropic effects (46), and indeed, our data suggest that this may have impacted performance. In particular, between conditions, participants reported no difference in their ratings of exertion across submaximal and maximal exercise, despite a lower absolute work output during the 20-min time trial following inhalation of THC-predominant cannabis. This is intriguing given that we received anecdotal reports from numerous participants describing the perceived benefits of cannabis during exercise and an expectation of improved performance. This finding does align with recently published data demonstrating that runners consuming cannabis report unchanged RPE but lower running speed (47). It should also be considered that the beneficial effects of THC on factors that hamper exercise tolerance such as pain or anxiety could be argued to lead to net improvements in performance, for example, in populations using THC medicinally. A separate observation from the evaluation of RPE is that CBD-predominant cannabis had no effect on perceived exertion, and power output was similar to control conditions. This finding aligns with the accepted notion that CBD has little to no intoxicating effect (48). Across all conditions, it should be noted that pacing strategies remained similar, suggesting that no intervention altered the strategic approach of the participants, despite the presence of clear psychotropic effects with some interventions. This is important in evaluating the performance effects of cannabis, given that deception and inhibition of neurophysiological feedback are able to alter pacing strategies and performance (49, 50).

The relevance of the magnitude change in time-trialing ability is worthy of consideration in the context of bona fide sports performance. First, the use of a 20-min time trial in this study represents only the second use of an ecologically valid (34, 51) performance test after cannabis use. Although not a direct analog of a competitive situation, this test has been demonstrated to strongly relate to functional threshold power (34), a widely used performance metric in cycling. Our data reveal that cannabis containing THC reduces performance by 5.1% when smoked and 7.5% when inhaled as an aerosol. This magnitude reduction in mean power output represents a highly impactful performance decrement given that within-athlete performance variability across cycling disciplines has been reported to range between 0.6% and 2.4% (52).

A number of important limitations of the present should be acknowledged. We did not measure blood levels of cannabinoids throughout our experiments owing to competing needs within a larger clinical trial. In combination with the fact that we did not assess intoxication and participants were ultimately able to select their own inhalation practices, we could not perfectly control the dose administered in each condition. Nevertheless, the substantial physiological and performance effects we observed led us to believe that our inhalation protocols were effective. Owing to concerns expressed by regulatory bodies, we were also required to perform the control visit before any cannabis interventions, making it impossible for us to fully randomize conditions. Although this could be argued to introduce a learning effect, it is logical to assume that participants would improve at the performance test with repeated attempts. In reality, we observed the highest power outputs during the 20-min time trial in our control visit, which could actually argue for an underestimation of the effects of THC-containing cannabis, and further support, rather than contradict our findings. Further along the lines of study design, the present study was not placebo controlled. Although a placebo control would be ideal, such an intervention remains elusive in cannabis research given the easily discernible psychotropic effects of THC, which in our experience, participants could immediately identify. Finally, our study lacks insight into a number of factors that could further help explain our findings. Pre-exercise performance expectations were not evaluated and could be argued to influence subsequent performance outcomes. However, our laboratory has specifically demonstrated that performance expectations do not always align with performance outcomes, and performance can even contradict negative expectations (53). There is also some evidence to suggest that tolerance to the cardiovascular effects of THC can occur with repeated use (54). Given that we collected only sufficient data to screen participants' cannabis use behaviors rather than comprehensively characterize them, it is difficult for us to comment on whether the effects we observed in our participants may have been blunted or accentuated, respectively, by either the presence of or lack of tolerance to the effects of THC. Finally, although we collected urine samples for drug screening, we did not assess hydration status. This might be considered a limitation in the context of the present findings given that our main finding of reduced performance on a 20-min time trial and elevated exercising heart rate at submaximal exercise both occur with dehydration and therefore could explain our results. However, we believe this to be highly improbable given that this situation would require incidences of dehydration to occur with systemically greater frequency specifically during THC conditions, and not control or CBD conditions.

There are a number of key findings from this investigation. First, cannabis containing THC alters the physiological responses to exercise, in particular, when it is consumed via smoking rather than aerosol inhalation. Concomitant with this altered response, cannabis containing THC detrimentally impacts cycling performance in a 20-min time trial, likely due to a combination of the physiological and psychotropic effects of cannabinoid consumption. In contrast,

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inhalation of CBD-predominant cannabis devoid of THC has little effect on exercise physiology and human performance. Each of these findings should be considered as athletes, coaches, and members of regulatory bodies in sports cope with the ever-changing landscape surrounding cannabis use both within and outside of sports, around the world. Finally, our data highlight the now urgent need for more research examining the physiological effects of cannabis itself and the interactions of THC and CBD on all systems and tissues involved in the oxygen cascade, particularly the heart and peripheral vasculature.

DATA AVAILABILITY

Data will be made available upon reasonable request.

SUPPLEMENTAL DATA

Supplemental Figure: https://doi.org/10.6084/m9.figshare.24424414.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

C.P.C. and J.F.B. conceived and designed research; C.P.C. and R.E.B. performed experiments; C.P.C. analyzed data; C.P.C. and J.F.B. interpreted results of experiments; C.P.C. prepared figures; C.P.C. drafted manuscript; C.P.C., R.E.B., A.M.C., and J.F.B. edited and revised manuscript; C.P.C., R.E.B., A.M.C., and J.F.B. approved final version of manuscript.

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